

ECON 400 | Term Paper

The Effect of Fast Food Consumption on Obesity Rates Across United States

Submitted by:

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ECON 400 | Econometrics


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Introduction

Obesity is one of the most pressing public health challenges in the United States. As of 2021, over one-third of American adults were classified as obese (BMI of 30 or higher), with rates varying considerably across states. The economic burden is substantial, as obesity-related medical costs are estimated at over \$170 billion annually. States with higher obesity rates face greater strain on healthcare systems, higher rates of chronic disease including type 2 diabetes and cardiovascular disease, and reduced workforce productivity.

This paper examines what drives these state-level differences using 2021 cross-sectional data and OLS regression. The explanatory variables include fast food restaurant density, median household income, physical inactivity rate, diagnosed diabetes rate, and urban classification. Understanding these determinants is relevant not only for public health policy but also for businesses in the food service and healthcare industries, which operate under increasing regulatory scrutiny as obesity costs continue to grow.

Data Sources

This study uses a cross-sectional dataset of 50 U.S. states for 2021, compiled from four public government sources. Florida was excluded because the state did not collect sufficient BRFSS responses in 2021 to meet the minimum inclusion requirement for the public-use dataset, resulting in a final sample of 50 observations including Washington D.C. Because this is a cross-sectional study analyzing variation across locations at a single point in time, the minor year differences between sources (2020 for fast food density, 2021 for health variables, 2023 for income) represent a data limitation but do not introduce time series concerns.

Dependent variable: Adult obesity rate (% of adults with BMI ≥ 30), sourced from the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2021 annual survey.

Explanatory variables:

- **FastFoodDensity:** fast food establishments per person by state (USDA Food Environment Atlas, 2020). *For Maine, New York, and Texas, the state-level value was suppressed; county averages were used as approximations.*
- **MedianHouseholdIncome:** median household income by state (Census Bureau ACS 2023, Table S1901).
- **PhysicalInactivity:** percentage of adults reporting no leisure-time physical activity, aggregated from county-level data (CDC USDSS, 2021).
- **DiabetesRate:** percentage of adults diagnosed with diabetes, aggregated from county-level data (CDC USDSS, 2021).
- **Urban:** binary dummy variable (1 = metro, 0 = non-metro), derived from METRO23 in the USDA Food Environment Atlas Socioeconomic tab.
- **FFD_Income:** interaction term (FastFoodDensity $\times$ MedianHouseholdIncome), constructed in R.

Models & Methods

OLS regression is applied to a cross-sectional sample of 50 U.S. states. The model is specified as:

$$\begin{aligned} ObesityRate = & \beta_0 + \beta_1(FastFoodDensity) + \beta_2(MedianHouseholdIncome) \\ & + \beta_3(PhysicalInactivity) + \beta_4(DiabetesRate) + \beta_5(Urban) \\ & + \beta_6(FFD_Income) + \varepsilon \end{aligned}$$

Individual coefficient significance is assessed using t-tests at the 5% level. The overall model is assessed using the F-statistic. The Breusch-Pagan test is applied to detect heteroskedasticity, with HC1 robust standard errors applied as treatment. VIF is used to diagnose multicollinearity, with $VIF > 10$ as the threshold. The interaction term FFD_Income tests whether the effect of fast food density on obesity differs across income levels.

Data Analysis

Descriptive Statistics

Table 1 summarizes descriptive statistics for all variables. The mean obesity rate across 50 states is 33.45%, ranging from 24.70% (D.C.) to 40.60% (West Virginia). Median household income averages \$77,674 with substantial variation across states (\$54,203 to \$108,210). Fast food density averages 0.72 establishments per person. Physical inactivity averages 20.20% and diagnosed diabetes averages 8.68%. Approximately 46% of states are classified as urban.

Correlation Analysis

The correlation matrix (Table 2) shows that median household income has the strongest relationship with obesity at -0.837, followed by physical inactivity at 0.753 and diabetes rate at 0.643. Fast food density shows a weak positive correlation of 0.232. Physical inactivity and diabetes rate are highly correlated with each other at 0.821, which may reduce the precision of individual coefficient estimates in the full model. Figures 2 and 3 illustrate the two strongest bivariate relationships.

OLS Regression Results

Table 3 presents the OLS regression results. The model yields an R^2 of 0.7964, indicating that approximately 80% of the variation in state-level obesity rates is explained by the included variables. The adjusted R^2 is 0.768 and the F-statistic is 28.03 ($p < 0.001$), confirming joint significance of the model.

Among individual coefficients, only MedianHouseholdIncome is statistically significant at the 5% level ($t = -2.594$, $p = 0.013$). The coefficient of -0.0002 implies that a \$1,000 increase in median household income is associated with a 0.20 percentage point decrease in the obesity rate, holding all else constant. This is consistent with the strong negative correlation observed in the descriptive analysis. FastFoodDensity, PhysicalInactivity, DiabetesRate, Urban, and the interaction term are not individually significant, likely due to the high intercorrelation between PhysicalInactivity and DiabetesRate ($r = 0.821$), which reduces the precision of each estimate.

Heteroskedasticity Test

The Breusch-Pagan test yields $BP = 7.184$ ($df = 6$, $p = 0.3042$). The null hypothesis of homoskedasticity is not rejected at the 5% level. There is no statistically significant evidence of heteroskedasticity. HC1 robust standard errors were applied regardless as a precautionary measure. Under robust standard errors, MedianHouseholdIncome remains significant at the 1% level ($t = -3.485$, $p = 0.001$), confirming the robustness of the key finding.

Robust Standard Errors

Table 5 presents coefficient estimates with HC1 robust standard errors. Point estimates are identical to Table 3 since robust standard errors adjust only the variance-covariance matrix. Under HC1, MedianHouseholdIncome becomes significant at the 1% level ($t = -3.485$, $p = 0.001$), strengthening from the 5% level under standard OLS. All other variables remain insignificant, confirming the baseline results are not driven by heteroskedasticity.

Multicollinearity Check (VIF)

VIF values were estimated on the model without the interaction term. All values fall below the threshold of 10 (Table 6). PhysicalInactivity has the highest VIF at 4.68, consistent with its correlation with DiabetesRate ($r = 0.821$), but does not indicate problematic multicollinearity. OLS estimates remain reliable.

Conclusions

This study examined the determinants of adult obesity rates across 50 U.S. states using OLS regression with 2021 data. The model explains approximately 80% of the variation in state-level obesity rates ($R^2 = 0.7964$, $F = 28.03^{***}$). Median household income is the only individually significant predictor, with a coefficient of -0.0002 ($p = 0.013$). Higher-income states consistently exhibit lower obesity rates. This finding holds under robust standard errors ($p = 0.001$).

Physical inactivity and diabetes rate showed strong bivariate correlations with obesity (0.753 and 0.643, respectively) but were not individually significant in the full model, likely due to their high intercorrelation ($r = 0.821$). Fast food density showed a weak and insignificant positive relationship when controlling for other factors. The Breusch-Pagan test confirmed no heteroskedasticity, and all VIF values were below the threshold of 10.

Limitations include the exclusion of Florida due to missing data, the use of county-level averages for Maine, New York, and Texas fast food density, and minor year mismatches between data sources (income from 2023, fast food density from 2020). The cross-sectional design limits causal inference, as unmeasured confounders such as cultural dietary habits or state-level public health investment may drive results.

From a policy standpoint, the findings suggest that income-related interventions, such as expanding access to affordable healthy food in lower-income states and subsidizing healthier food alternatives, are likely to have the greatest impact on reducing obesity. Businesses in the food service and healthcare industries operating in lower-income states should anticipate continued regulatory pressure as obesity-related healthcare costs remain a growing concern.

References

U.S. Department of Agriculture, Economic Research Service. (2025). *Food Environment Atlas*.

<https://www.ers.usda.gov/data-products/food-environment-atlas/data-access-and-documentation-downloads/>

U.S. Census Bureau. (2023). *American Community Survey 1-Year Estimates: Income in the past 12 months, Table S1901*. <https://data.census.gov/table/ACSST1Y2023.S1901>

Centers for Disease Control and Prevention. (2021). *Behavioral Risk Factor Surveillance System: Adult obesity prevalence by state*. https://www.cdc.gov/brfss/annual_data/annual_data.htm

Centers for Disease Control and Prevention, United States Diabetes Surveillance System. (2021). *Physical inactivity by county*. <https://usdss.cdc.gov/diabetes/analysis.html>

Centers for Disease Control and Prevention, United States Diabetes Surveillance System. (2021). *Diagnosed diabetes by county*. <https://usdss.cdc.gov/diabetes/analysis.html>

Appendix

Table 1: Descriptive Statistics

Variable	n	Mean	SD	Min	Max	Skew
ObesityRate (%)	50	33.45	4.09	24.70	40.60	-0.25
FastFoodDensity	50	0.72	0.22	0.30	1.16	0.12
MedianIncome (\$)	50	77,674	13,019	54,203	108,210	0.33
PhysicalInactivity (%)	50	20.20	2.39	15.20	24.93	0.07
DiabetesRate (%)	50	8.68	1.03	6.95	11.25	0.62
Urban (dummy)	50	0.46	0.50	0	1	0.16

Figure 1: Adult obesity Rate by State (Ranked)

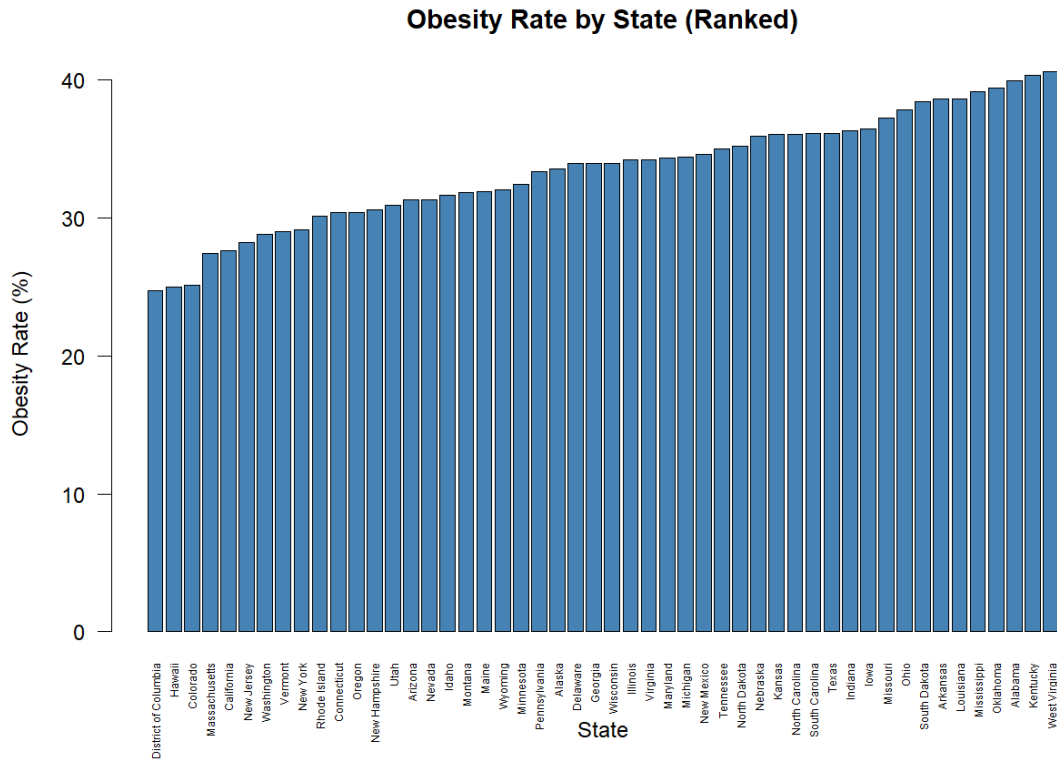


Table 2: Correlation Matrix

Variable	Obesity	FastFood	Income	Inactivity	Diabetes	Urban
ObesityRate	1.000	0.232	-0.837	0.753	0.643	-0.260
FastFoodDensity	0.232	1.000	-0.099	0.143	0.142	0.022
MedianIncome	-0.837	-0.099	1.000	-0.674	-0.548	0.108
Physicallnactivity	0.753	0.143	-0.674	1.000	0.821	-0.323
DiabetesRate	0.643	0.142	-0.548	0.821	1.000	-0.138
Urban	-0.260	0.022	0.108	-0.323	-0.138	1.000

Figure 2: Obesity Rate vs Median Household Income

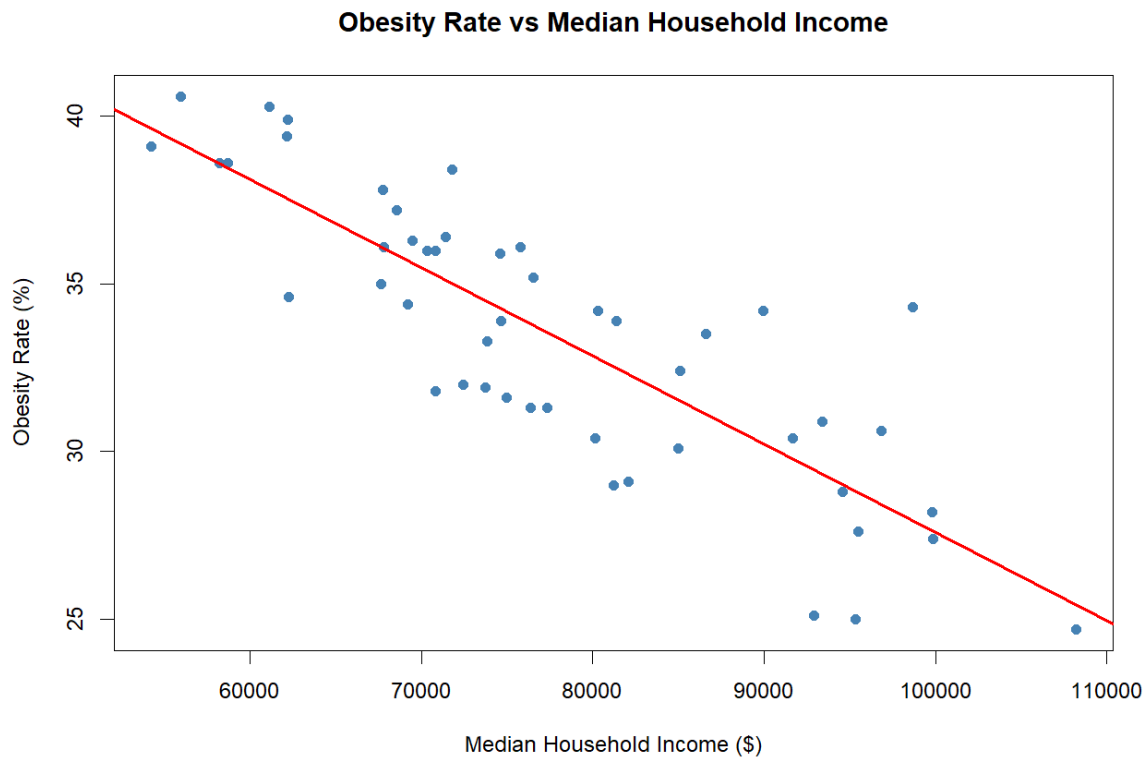


Figure 3: Obesity Rate vs Physical Inactivity Rate

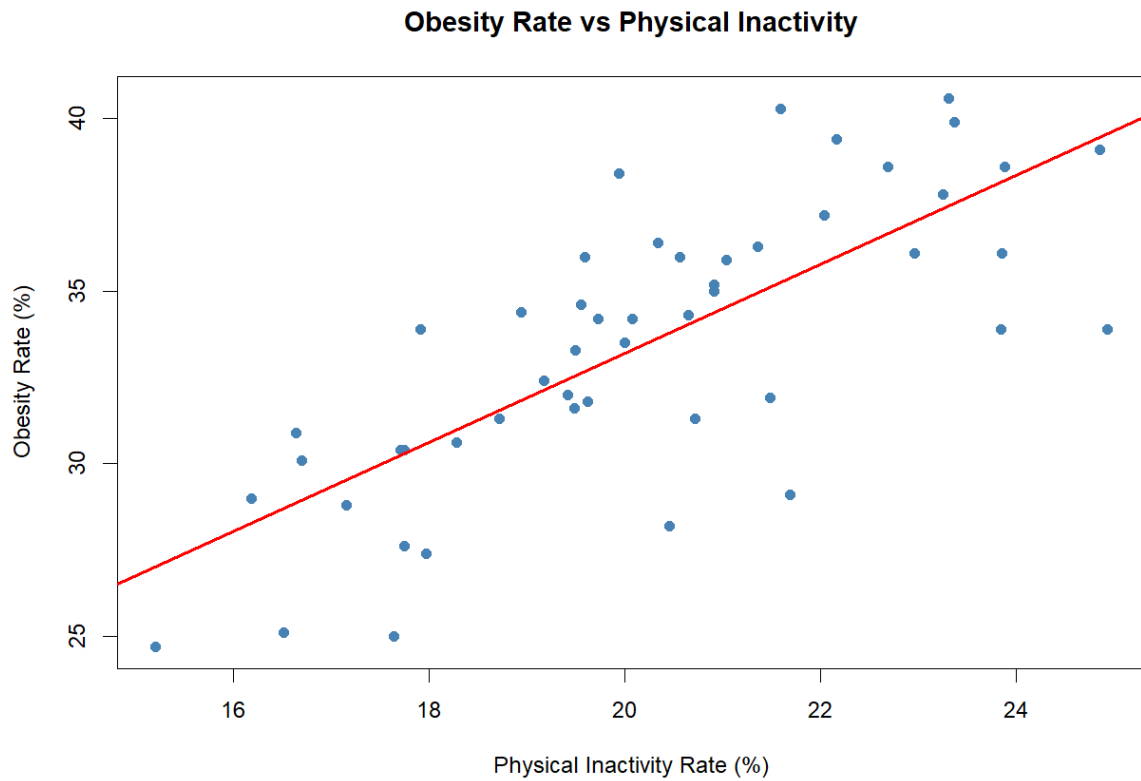


Table 3: OLS Regression Results

Variable	Estimate	Std. Error	t-value	p-value
(Intercept)	38.1281	7.8808	4.838	<0.001 ***
FastFoodDensity	1.4251	8.5506	0.167	0.8684
MedianHouseholdIncome	-0.0002	0.0001	-2.594	0.0129 *
PhysicalInactivity	0.2948	0.2605	1.132	0.2641
DiabetesRate	0.4776	0.5014	0.952	0.3462

Urban	-1.0019	0.6183	-1.621	0.1124
FFD_Income (interaction)	0.0000	0.0001	0.124	0.9022
R² = 0.7964 Adj. R² = 0.768 F = 28.03*** n = 50				

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 4: Heteroskedasticity Tests

Test	BP Statistic	df	p-value	Conclusion
Breusch-Pagan	7.184	6	0.3042	Fail to reject H0, no HSK

Table 5: Regression Results with HC1 Robust Standard Errors

Variable	Estimate	Robust SE	t-value	p-value
(Intercept)	38.1280	6.4243	5.935	<0.001 ***
FastFoodDensity	1.4251	6.9338	0.206	0.8381
MedianHouseholdIncome	-0.0002	0.0001	-3.485	0.0011 **
PhysicalInactivity	0.2948	0.2616	1.127	0.2660
DiabetesRate	0.4776	0.4852	0.984	0.3305
Urban	-1.0019	0.6329	-1.583	0.1207
FFD_Income (interaction)	0.0000	0.0001	0.153	0.8788

Table 6: Multicollinearity Check (VIF)

Variable	VIF	Conclusion
FastFoodDensity	1.03	No concern
MedianHouseholdIncome	1.88	No concern
PhysicalInactivity	4.68	Moderate, below threshold
DiabetesRate	3.25	No concern
Urban	1.22	No concern

```

3 # -----
4 # ECON 400 Term Paper: R code
5 # Carolina Varanko
6 # -----
7
8 library(lmtest)
9 library(sandwich)
10 library(car)
11 library(psych)
12
13 # ---- LOAD DATA + CREATE INTERACTION TERM ----
14 data <- read.csv("dataset_termpaper_cvaranko.csv")
15 head(data)
16 str(data)
17
18 data$FFD_Income <- data$FastFoodDensity * data$MedianHouseholdIncome
19
20 # ---- DESCRIPTIVE STATISTICS ----
21 desc_vars <- data[, c("ObesityRate", "FastFoodDensity", "MedianHouseholdIncome",
22                      "PhysicalInactivity", "DiabetesRate", "Urban")]
23 describe(desc_vars)
24
25 # ---- CORRELATION MATRIX ----
26 round(cor(desc_vars, use = "complete.obs"), 3)
27
28 # ---- OLS REGRESSION ----
29 model <- lm(ObesityRate ~ FastFoodDensity + MedianHouseholdIncome +
30           PhysicalInactivity + DiabetesRate + Urban + FFD_Income,
31           data = data)
32 summary(model)
33
34 # ---- BREUSCH-PAGAN TEST ----
35 # If p-value < 0.05, heteroskedasticity is present
36 bptest(model)
37
38 # ---- ROBUST STANDARD ERRORS ----
39 coeftest(model, vcov = vcovHC(model, type = "HCl"))
40
41 # ---- MULTICOLLINEARITY CHECK (VIF) ----
42 # VIF > 10 indicates problematic multicollinearity
43 model_no_interact <- lm(ObesityRate ~ FastFoodDensity + MedianHouseholdIncome +
44                       PhysicalInactivity + DiabetesRate + Urban,
45                       data = data)
46 vif(model_no_interact)
47
48 # ---- COEFFICIENT SUMMARY ----
49 # Model: ObesityRate ~ FastFoodDensity + MedianHouseholdIncome + PhysicalInactivity + DiabetesRate + Urban + FFD_Income
50 round(summary(model)$coefficients, 4)
51 round(summary(model)$r.squared, 4)
52 round(summary(model)$adj.r.squared, 4)
53 round(summary(model)$fstatistic[1], 4)
54 nobs(model)
55
56 # ---- VISUALIZATIONS ----
57
58 # Plot 1: Obesity Rate vs Median Household Income
59 plot(data$MedianHouseholdIncome, data$ObesityRate,
60      main = "Obesity Rate vs Median Household Income",
61      xlab = "Median Household Income ($)",
62      ylab = "Obesity Rate (%)",
63      pch = 16, col = "steelblue")
64 abline(lm(ObesityRate ~ MedianHouseholdIncome, data = data),
65       col = "red", lwd = 2)
66
67 # Plot 2: Obesity Rate by State (ranked bar chart)
68 sorted_data <- data[order(data$ObesityRate), ]
69 barplot(sorted_data$ObesityRate,
70        names.arg = sorted_data$State,
71        main = "Obesity Rate by State (Ranked)",
72        xlab = "State",
73        ylab = "Obesity Rate (%)",
74        col = "steelblue",
75        las = 2,
76        cex.names = 0.5)
77
78 # Plot 3: Obesity Rate vs Physical Inactivity
79 plot(data$PhysicalInactivity, data$ObesityRate,
80      main = "Obesity Rate vs Physical Inactivity",
81      xlab = "Physical Inactivity Rate (%)",
82      ylab = "Obesity Rate (%)",
83      pch = 16, col = "steelblue")
84 abline(lm(ObesityRate ~ PhysicalInactivity, data = data),
85       col = "red", lwd = 2)
86

```

```

> data <- read.csv("dataset_termpaper_cvaranko.csv")
> head(data)
  State ObesityRate MedianHouseholdIncome FastFoodDensity Urban PhysicalInactivity DiabetesRate
1  Alabama      39.9           62212      0.7228200      0      23.37      10.50
2  Alaska       33.5           86631      0.8014961      1      20.00      8.70
3  Arizona      31.3           77315      0.7501516      1      20.72      9.05
4  Arkansas     38.6           58700      0.9760462      0      23.88      9.78
5  California   27.6           95521      0.3162270      1      17.75      8.16
6  Colorado     25.1           92911      0.4146582      1      16.51      7.22
> str(data)
'data.frame':  50 obs. of  7 variables:
 $ State      : chr  "Alabama" "Alaska" "Arizona" "Arkansas" ...
 $ ObesityRate : num  39.9 33.5 31.3 38.6 27.6 25.1 30.4 33.9 24.7 33.9 ...
 $ MedianHouseholdIncome: int  62212 86631 77315 58700 95521 92911 91665 81361 108210 74632 ...
 $ FastFoodDensity : num  0.723 0.801 0.75 0.976 0.316 ...
 $ Urban         : int  0 1 1 0 1 1 0 0 1 0 ...
 $ PhysicalInactivity : num  23.4 20 20.7 23.9 17.8 ...
 $ DiabetesRate  : num  10.5 8.7 9.05 9.78 8.16 ...

> describe(desc_vars)
      vars  n   mean    sd  median trimmed   mad   min   max  range  skew kurtosis
ObesityRate      1  50  33.45   4.09   33.90   33.59   3.85  24.70  40.60  15.90 -0.25   -0.67
FastFoodDensity   2  50   0.72   0.22   0.71   0.72   0.25   0.30   1.16   0.86  0.12   -0.85
MedianHouseholdIncome 3  50 77674.46 13019.18 74787.00 77304.00 10507.19 54203.00 108210.00 54007.00 0.33  -0.74
PhysicalInactivity 4  50  20.20   2.39   20.04   20.18   2.53  15.20  24.93   9.73  0.07  -0.77
DiabetesRate      5  50   8.68   1.03   8.39   8.60   0.96   6.95  11.25   4.30  0.62  -0.30
Urban            6  50   0.46   0.50   0.00   0.45   0.00   0.00   1.00   1.00  0.16  -2.01
      se
ObesityRate      0.58
FastFoodDensity   0.03
MedianHouseholdIncome 1841.19
PhysicalInactivity 0.34
DiabetesRate      0.15
Urban            0.07

> round(cor(desc_vars, use = "complete.obs"), 3)
      ObesityRate FastFoodDensity MedianHouseholdIncome PhysicalInactivity DiabetesRate Urban
ObesityRate      1.000      0.232      -0.837      0.753      0.643 -0.260
FastFoodDensity   0.232      1.000      -0.099      0.143      0.142  0.022
MedianHouseholdIncome -0.837      -0.099      1.000      -0.674      -0.548  0.108
PhysicalInactivity   0.753      0.143      -0.674      1.000      0.821 -0.323
DiabetesRate        0.643      0.142      -0.548      0.821      1.000 -0.138
Urban              -0.260      0.022      0.108      -0.323      -0.138 1.000

> summary(model)

Call:
lm(formula = ObesityRate ~ FastFoodDensity + MedianHouseholdIncome +
    PhysicalInactivity + DiabetesRate + Urban + FFD_Income, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-3.6564 -1.6672 -0.0059  1.5039  4.0956

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.813e+01  7.881e+00   4.838 1.72e-05 ***
FastFoodDensity  1.425e+00  8.551e+00   0.167  0.8684
MedianHouseholdIncome -2.069e-04  7.977e-05 -2.594  0.0129 *
PhysicalInactivity  2.948e-01  2.605e-01   1.132  0.2641
DiabetesRate    4.776e-01  5.014e-01   0.952  0.3462
Urban          -1.002e+00  6.183e-01  -1.621  0.1124
FFD_Income      1.311e-05  1.060e-04   0.124  0.9022
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.97 on 43 degrees of freedom
Multiple R-squared:  0.7964, Adjusted R-squared:  0.768
F-statistic: 28.03 on 6 and 43 DF, p-value: 2.363e-13

```



```

> bptest(model)

studentized Breusch-Pagan test

data: model
BP = 7.184, df = 6, p-value = 0.3042

> coeftest(model, vcov = vcovHC(model, type = "HC1"))

t test of coefficients:

              Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.8128e+01 6.4243e+00  5.9350 4.546e-07 ***
FastFoodDensity  1.4251e+00 6.9338e+00  0.2055 0.838129
MedianHouseholdIncome -2.0691e-04 5.9380e-05 -3.4845 0.001147 **
PhysicalInactivity  2.9480e-01 2.6157e-01  1.1270 0.265979
DiabetesRate    4.7755e-01 4.8522e-01  0.9842 0.330524
Urban          -1.0019e+00 6.3290e-01 -1.5831 0.120726
FFD_Income     1.3110e-05 8.5454e-05  0.1534 0.878785
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> vif(model_no_interact)
FastFoodDensity MedianHouseholdIncome PhysicalInactivity DiabetesRate Urban
1.027279          1.882110          4.680347          3.249722 1.219156

> round(summary(model)$coefficients, 4)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  38.1281      7.8808  4.8381  0.0000
FastFoodDensity  1.4251      8.5506  0.1667  0.8684
MedianHouseholdIncome -0.0002      0.0001 -2.5939  0.0129
PhysicalInactivity  0.2948      0.2605  1.1315  0.2641
DiabetesRate    0.4776      0.5014  0.9523  0.3462
Urban          -1.0019      0.6183 -1.6205  0.1124
FFD_Income     0.0000      0.0001  0.1237  0.9022
> round(summary(model)$r.squared, 4)
[1] 0.7964
> round(summary(model)$adj.r.squared, 4)
[1] 0.768
> round(summary(model)$fstatistic[1], 4)
value
28.0315
> nobs(model)
[1] 50

```